

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100 Marks

Class: TE

Name of the Course: OPERATING SYSTEMS

Course Code: UCEC502

Duration: 3 Hours.

Semester: V

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks																		
Q 1 (a)	Explain the concept of kernel and shell in an operating system.	10																		
Q 1 (b)	Explain any five system calls with appropriate syntaxes.	05																		
Q2 (a)	Describe the differences among short term, medium term and long term scheduling.	10																		
Q2 (b)	What is a Semaphore? Explain in detail the usage of semaphore for resource acquisition and release.	10																		
Q2 (c)	Explain process creation and process termination? OR Consider the following snapshot of the processes with the length of CPU burst given in milliseconds. <table border="1"> <thead> <tr> <th>Process</th><th>Arrival time</th><th>Burst time</th></tr> </thead> <tbody> <tr> <td>P1</td><td>0</td><td>4</td></tr> <tr> <td>P2</td><td>2</td><td>5</td></tr> <tr> <td>P3</td><td>4</td><td>6</td></tr> <tr> <td>P4</td><td>5</td><td>2</td></tr> <tr> <td>P5</td><td>6</td><td>1</td></tr> </tbody> </table> Draw Gantt charts and calculate average turnaround time and average waiting time for the following scheduling algorithms: FCFS, SJF (nonpreemptive) and Round Robin (quantum = 2).	Process	Arrival time	Burst time	P1	0	4	P2	2	5	P3	4	6	P4	5	2	P5	6	1	10
Process	Arrival time	Burst time																		
P1	0	4																		
P2	2	5																		
P3	4	6																		
P4	5	2																		
P5	6	1																		

Q3 (a)	Consider the following snapshot of the system and Answer the following questions using Banker's algorithm: i) Determine the total amount of resources of each type. ii) What is the content of need matrix? iii) Determine if the system is in safe state using safety algorithm. iv) If a request from process p1 arrives for (0, 2, 2, 0) can the request be granted immediately. <table><tr><th rowspan="2">Process</th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P0</td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>7</td><td>5</td><td>0</td><td colspan="4" rowspan="4"></td></tr><tr><td>P2</td><td>1</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td><td>6</td></tr><tr><td>P3</td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td></tr><tr><td>P4</td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td></tr></table>	Process	Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P0	0	0	1	2	0	0	1	2	1	5	2	0	P1	1	0	0	0	1	7	5	0					P2	1	3	5	4	2	3	5	6	P3	0	6	3	2	0	6	5	2	P4	0	0	1	4	0	6	5	6	10
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Q3 (b)	Define the notion of a deadlock. Explain necessary and sufficient conditions for a deadlock to occur.	05																																																																														
Q4 (a)	On a disk with 1000 cylinders, numbers 0 to 999, compute the number of tracks the disk arm must move to satisfy all the requests in the disk queue. Assume the last request serviced was at track 345 and the head is moving toward track 0. The queue in FIFO order contains requests for the following tracks: 123, 874, 692, 475, 105, 376. Perform the computation for the following scheduling algorithms. (i) FIFO (ii) SSTF (iii)SCAN (iv)LOOK (v) C-SCAN	10																																																																														
Q4 (b)	Draw and explain address translation in a Segmentation/Paging system. OR Calculate Hit and Miss using LRU, Optimal and FIFO page replacement policies for the following sequence. Page frame size is 3. 0,4,3,2,1,4,6,3,0,8,9,3,8,5	10																																																																														
Q5 (a)	With the help of a diagram, explain UNIX file management. OR Draw and Explain UNIX process state transition diagram.	10																																																																														
Q5 (b)	Explain device I/O in Linux. OR Write a short note on Windows 7 operating system.	10																																																																														